IP Credit: Integrative Process

This credit applies to

- BD+C: New Construction (1 point)
- BD+C: Core and Shell (1 point)
- BD+C: Schools (1 point)
- BD+C: Retail (1 point)
- BD+C: Data Centers (1 point)
- BD+C: Warehouses and Distribution Centers (1 point)
- BD+C: Hospitality (1 point)
- BD+C: Healthcare (1 point)

INTENT

To support high-performance, cost-effective, equitable project outcomes through an early analysis of the interrelationships among systems.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Beginning in pre-design and continuing throughout the design phases, identify and use opportunities to achieve synergies across disciplines and building systems. Use the analyses described below to inform the owner's project requirements (OPR), basis of design (BOD), design documents, and construction documents.

Discovery:
Choose two of the following to analyze:

Energy-Related Systems

Establish an energy performance target (EUI) no later than the schematic design phase. The target must be established using one of the following metrics:

- kBtu per square foot-year (kWh per square meter-year) of site energy use
- KBtu per square foot-year (kWh per square meter-year) of source energy use
- pounds per square foot-year (Kg per square meter-year) of greenhouse gas emissions
- energy cost per square foot-year (cost per square meter-year)

Perform a preliminary "simple box" energy modeling analysis before the completion of schematic design that explores how to reduce energy loads in the building and accomplish related sustainability goals by questioning default assumptions. Assess strategies associated with each of the following, as applicable:

- *Site conditions.* Assess shading, exterior lighting, hardscape, landscaping, and adjacent site conditions.
- *Massing and orientation.* Assess how massing and orientation affect HVAC sizing, energy consumption, lighting, and renewable energy opportunities.
- *Basic envelope attributes.* Assess insulation values, window-to-wall ratios, glazing characteristics, shading, and window operability.
- *Lighting levels.* Assess interior surface reflectance values and lighting levels in occupied spaces.
- *Thermal comfort ranges.* Assess thermal comfort range options.
- *Plug and process load needs.* Assess reducing plug and process loads through programmatic solutions (e.g., equipment and purchasing policies, layout options).

- *Programmatic and operational parameters.* Assess multifunctioning spaces, operating schedules, space allotment per person, teleworking, reduction of building area, and anticipated operations and maintenance.

Water-Related Systems

Perform a preliminary water budget analysis before the completion of schematic design that explores how to reduce potable water loads in the building, reduce the burden on municipal supply or wastewater treatment systems, and accomplish related sustainability goals. Assess and estimate the project's potential nonpotable water supply sources and water demand volumes, including the following, as applicable:

- *Indoor water demand.* Assess flow and flush fixture design case demand volumes, calculated in accordance with WE Prerequisite Indoor Water Use Reduction.
- *Outdoor water demand.* Assess landscape irrigation design case demand volume calculated in accordance with WE Credit Outdoor Water-Use Reduction.
- *Process water demand.* Assess kitchen, laundry, cooling tower, and other equipment demand volumes, as applicable.
- *Supply sources.* Assess all potential nonpotable water supply source volumes, such as on-site rainwater and graywater, municipally supplied nonpotable water, and HVAC equipment condensate. Analyze how nonpotable water supply sources can contribute to the water demand components listed above.

Assessment for Resilience

Conduct a risk assessment of any identified natural or environmental hazards affecting the project site(s) and building function out of the following:

- Sea Level Rise and Storm Surge
- Flooding
- Hurricane and High Winds
- Earthquake
- Wildfire
- Drought
- Landslides
- Extreme Heat
- Winter Storms

Demonstrate how the risk assessment influenced the project design and enhanced the project's resilience to natural disasters, disturbances, and changing climate conditions; provide reasoning for not addressing any identified hazards.

Social Equity

Beginning in pre-design and continuing throughout the design phases, review and then complete the LEED Project Team Checklist for Social Impact in order to assess and select strategies to address issues of inequity within the project and its community, team and supply chain. Through research and consultation with key stakeholders, ensure that all responses within the Checklist are ultimately documented as "Yes" or "No," and complete all sections for Stakeholders and Goals.

Health & Well-being

Beginning in pre-design and continuing throughout the design phases, use the following steps to inform the design and construction documents:

- Establish health goals. Set clear and specific goals to promote the health of core groups, including:

- Building occupants and users
- Surrounding community
- Supply chain

Develop a statement of health goals for each population, including a summary of how this health goal relates to the highest priority health need for each population.

- Prioritize design strategies. Select specific design and/or programming strategies to address the project's health goals. This could be accomplished by holding a stand-alone "health charrette" or by integrating health considerations into an existing green charrette.
- Anticipate outcomes. Identify expected impacts on population health behaviors and outcomes associated with the project's prioritized design strategies.

Implementation:

Develop a Project Team Letter. Provide a dated letter on the letterhead of the Integrative Process Facilitator that summarizes the team's integrative process approach and describes the difference that this integrative approach made in terms of improving project team interaction and project performance.

- Describe the approach developed by the project team for engaging a clearly defined and manageable integrative design process beginning in pre-design and continuing throughout the design phases.
- The letter must include a separate summary for each issue area analyzed by the project team, describing how the analysis informed the design and building form decisions in the project's OPR and BOD and the eventual design of the project. Describe the most important goals for each issue area and provide clear guidance on how to evaluate the project's impact on the selected goals.

The creation of this letter should be a team effort facilitated by the Integrative Process Facilitator. The letter must be signed by all principal project team members and made available to key stakeholders including, but not limited to the owner(s), facility manager(s), tenant(s), and community members. Describe how the letter was distributed to these stakeholders and/or made publicly available.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Refer to the LEED v4 reference guide for an introduction to the integrative process.

More than ever, the Integrative Process credit documents the nature of the process, the understanding of system relationships, and the resultant decision making by all project team members through a project team letter. Project teams are better able to demonstrate the difference between the standard approach and the integrative approach for key issue areas like energy and water, as well as broader concepts at the frontier of the green building movement like site selection, social equity, and health and well-being.

Step-by-Step Guidance

Follow steps 1-7 in the LEED v4 reference guide, with the following modifications:

- All references to the Integrative Process Worksheet are replaced by the Project Team Letter.
- Assess strategies associated with each of the seven energy aspects, as applicable, and at least one on-site non-potable water source that could supply a portion of at least two water demand components.
- Step 1: Add the following paragraph at the end:

Consider reviewing ASHRAE Standard 209-2018, which provides a standardized methodology for applying energy modeling throughout the integrative design process to inform building design.

- Step 2: Add the following paragraph at the end:
  ASHRAE 209, Section 5.3 (Climate and Site Analysis) and Section 5.4 (Benchmarking) provide helpful guidance for conducting this preliminary research.
- Step 4: Add the following paragraph at the end:
  ASHRAE Standard 209-2018 Section 5.5 (Energy Charrette) provides a good framework for incorporating energy considerations into the design charrette.
- Step 5:
  - Replace last sentence beginning with "Conduct" with "Conduct such preliminary modeling to assess at least one strategy for each of the above seven aspects, as applicable."
  - Add the following paragraph at the end:
    ASHRAE Standard 209-2018 Sections 6.1 (Simple Box Modeling), 6.2 (Conceptual Design Modeling), and Modeling Cycle 3 (Load Reduction Modeling) may be used to demonstrate compliance with the Integrative Process credit requirements to develop a Simple Box Energy Model. The data reporting information described in Standard 209, Section 5.7 may also be used to generate a consistent reporting methodology during the energy analysis process.
- Step 7: Replace" "…identify at least two options for each of the seven aspects listed in Step 5" with "…identify one or more options for each of the seven aspects listed in Step 5, as applicable."

Further Explanation

Refer to the LEED v4 reference guide, with the following additions and modification:

Project teams may choose additional lenses through which to demonstrate the outcomes and benefits of an integrative process which include site selection, social equity, and/or health & well-being.

Required Documentation

- Project Team Letter

Referenced Standards

- ASHRAE Standard 209-2018, Energy Simulation Aided Design for Buildings except Low Rise Residential Buildings

Connection to Ongoing Performance

- LEED O+M IN credit Innovation: The final phase of the integrative process is the period of occupancy, operations, and performance feedback. Project teams can demonstrate their ongoing efforts in the LEED v4.1 O+M Integrative Process pilot credit.